PRODUCT REQUIREMENTS DOCUMENT · V3.0 · STATUS & SUBMISSION GAP ANALYSIS

MPSTME GUARDIAN

What is built, what is left, and everything still owed for the course faculty's MDRIIA Capstone submission

MuJoCo · Dual-UAV Surveillance · MDRIIA Group 05

> *"To what extent can a collaborative dual-UAV surveillance system simulated in MuJoCo optimize campus perimeter patrol cycle time and OpenCV human detection latency compared to static security guard patrols?"*

COURSE	MDRIIA — Modern Day Robotics & Its Industrial Applications (702CO0E012) · Capstone ICA, 20 marks
FACULTY	The course faculty, SVKM's NMIMS MPSTME, Mumbai
GROUP	05 — E043 Muaaz Shaikh (Lead, owner of `main`) · E051 Aditya Rajkumar · E075 Zaid Rezaur Rahman · E077 Soumya Subhankar Ranasingh
STATUS AS OF	Thursday 29 October 2026 — **Dossier deadline (Mon 26 Oct, 23:59 IST) has passed · Expo & Viva: Saturday 31 October 2026**
REQUIREMENT SOURCES	Grading Rubric & Evaluation Policy · LinkedIn & GitHub Showcase Guide · Group 05 Technical Dossier template · Academic & Industry Standards Manual · Contributing / Branching guides · Group 05 scaffold README (course monorepo, read 29 Oct 2026)
SUPERSEDES	PRD v2.0 (8 Oct 2026, exterior scene) for status and submission scope; v2.0 remains the design reference

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LEGEND. DONE implemented and checked · PARTIAL exists but does not yet meet the requirement · NOT STARTED nothing usable yet · DECISION needs a ruling (team or faculty). Priorities: **P0** = marks are lost or the submission is invalid without it · **P1** = needed for full marks in a criterion · **P2** = quality / viva safety.

1 Executive Summary

The engineering core — a MuJoCo model of the NMIMS Main Building, two flying quadrotors with a cascaded controller, collision-safe coordination and a provenance-gated analytics pipeline — is built and working. What is missing is almost everything the rubric actually *grades* as evidence: the 500 Hz real-trial dataset, the statistics, the CSBS business model, the three teammates' code regions, the LinkedIn video and the filled Technical Dossier. The dossier and video deadlines have already passed.

2 days
TO EXPO & VIVA (SAT 31 OCT)

3 days
DOSSIER OVERDUE (DUE 26 OCT)

1 / 4
STUDENT CODE REGIONS DONE (E043 ONLY)

0
REAL TRIALS LOGGED TO `*_REAL_TRIALS.CSV`

0 / 6
DOSSIER SECTIONS FILLED

The five things that decide the marks

#	ITEM	MARKS IT FEEDS	STATUS	OWNER
1	Filled Technical Dossier (.docx + .pdf) committed to `docs/` and turned in on MS Teams by **each** member	All 20 (no turn-in ⇒ no marks for that student)	NOT STARTED	All · E043 compiles
2	N ≥ 30 (manual: ≥ 50) seeded MuJoCo runs per condition, 500 Hz, in `analytics/campus_patrol_real_trials.csv`, with Welch t-test, Cohen's d, 95 % CI	C1 Telemetry 2 + Statistics 2	PARTIAL	E043 + E075
3	CSBS script computing H_reclaimed, P_payback, NCRI/κ, M/M/c delay, FMEA, sensitivity graph — no hard-coded results, no currency	C2 — all 5	NOT STARTED	E077
4	Surveillance FSM (PATROL → DETECT → VERIFY → ALERT → TRACK → RETURN) with E051 detector and E075 geofence implemented	C1 Control 2 + viva for E051/E075	PARTIAL	E051, E075, E043
5	3–4 min LinkedIn video (live sim, business graph, all four on webcam), AI audit ×3, peer review of Groups 06 & 11	C3 — 2 of 5	NOT STARTED	All

IMMEDIATE ACTION (TONIGHT): message the course faculty that the dossier is late, state the new turn-in time, and ask the four questions in §15 (late acceptance, indoor scene vs "perimeter" title, N = 30 vs 50, OpenCV vs YOLO). Then every member works their P0 list in §6.

2 What Sir Requires — the Submission Contract

Consolidated from the Grading Rubric, the Showcase Guide and the Dossier template. These are the deliverables that are *checked*; anything not listed here earns no marks on its own.

#	DELIVERABLE	WHERE IT GOES	DEADLINE	STATUS
D1	Technical Dossier completed in **Microsoft Word desktop** (drop-downs and "type here" fields kept intact, not Google Docs), exported to PDF	`Group_05_…/docs/MDRIIA_Group_05_Technical_Dossier.docx` **and** `.pdf`, original file name	Mon 26 Oct 23:59 — **passed**	NOT STARTED
D2	Dossier PDF attached and **"Turn in" clicked individually** by all four members	MS Teams assignment	Mon 26 Oct — **passed**	NOT STARTED
D3	Group folder in the **course monorepo**: MJCF model, controller, business script, `*_real_trials.csv`	`github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026` via PR into `main`	With D1	PARTIAL — work is only in the private `MuaazSM/mdriia-project`
D4	3:00–4:00 LinkedIn video, uploaded natively, one post per group, teammates + MPSTME + the course faculty tagged, monorepo folder linked	LinkedIn; URL in dossier §5 and `docs/TEAM_ROSTER.json → linkedin_url`	Sun 25 Oct 18:00 — **passed**	NOT STARTED
D5	Peer review of Group 06 and Group 11 videos (0–2 on Problem, Physics, Business, Data, Clarity + best moment + one viva question)	Dossier §5.2	With D1	NOT STARTED
D6	Live Expo demo: simulation running from a **fresh `git clone` + `pip install -r requirements.txt`**, FSM states visible, one handled failure shown; business graphs; webcam PiP of every member; **no PPT**	Studio Showcase	Sat 31 Oct	PARTIAL
D7	Individual viva: own role and code *and* the whole project including the business model	Viva voce	Sat 31 Oct	PARTIAL

Rules that can zero a component

RULE	WHAT IT MEANS FOR US	STATUS
Data provenance — `SIMULATED_*` and starter figures are `np.random` targets	The scaffold CSV (still named `campus_patrol_benchmark.csv` in our repo) and the three starter PNGs must never be reported. Our figure script already refuses them.	PARTIAL rename pending
No hard-coded results in the business script	`campus_security_economics.py` still contains 48.72, 7.84, 72.0, 50.4, 0.66, 0.31, 35.8, 1.2 as constants	VIOLATING
Zero Currency Rule — ratios only (κ, NCRI, P_payback)	E077's branch economics carries a currency code field	VIOLATING (branch)
Every claim/citation verified at its primary source (§1.1)	72 % latency cut, 11-month payback, 90-s verification, "45 → 8.5 min" are untraceable	NOT STARTED
AI use logged (§5.1); undisclosed AI content = integrity breach	AI was used heavily (scene, controller, documents, E077 runtime). Must be declared in §5.1 and §6.	NOT STARTED
Attribution matches git history (checked with `git shortlog -sn -- Group_05_*/`)	Each member must commit their own work from their own account; ≥ 4 feature commits each (Standards Manual §5)	AT RISK

3 Marks Map: 20 Marks, Criterion by Criterion

Component 1 — CDIO Robotics Engineering & MuJoCo Physics (10)

CRITERION	MK	REQUIRED EVIDENCE	WHERE WE ARE	STATUS
Problem framing	2	Authorised title; verified baseline friction; governing standards with clauses; claim verification table (§1)	Title fixed. No baseline metric with source; no standards decisions; three seeded claims unverified.	NOT STARTED
MuJoCo rigid-body model	2	`models/*.xml` loads; friction / `solref` / timestep justified; mechanism matches title	Loads (interior + exterior). Timestep **0.004 s** (sir wants 0.002 s). Friction 1.0/0.01/0.005 and solref 0.004 1 not yet justified in writing. Indoor scene vs “perimeter” wording — §15.	PARTIAL
Closed-loop control	2	Controller with implemented FSM and stated control law; honest authorship table	Cascaded P → PI → PD control law implemented. FSM covers flight phases only (takeoff → transit → descend → patrol); surveillance FSM blocked on E051/E075. Authorship table empty.	PARTIAL
Telemetry & Monte Carlo	2	500 Hz logging; N ≥ 30 randomised runs per condition, seeds recorded; tracking RMSE; control effort $\int \tau^2\,dt$	Seeded runner with provenance exists. Logging is once per outer step (≈ 30 Hz), not 500 Hz. RMSE and $\int \tau^2\,dt$ not computed. No batch run.	PARTIAL
Statistical validation	2	Baseline vs proposed; t-test; Cohen’s d; 95 % CI; plots regenerate from `*_real_trials.csv`	Baseline condition not defined; no statistics script; figures pending data.	NOT STARTED

Component 2 — CSBS Techno-Economic Business Systems (5) — owner E077

CRITERION	MK	REQUIRED EVIDENCE	WHERE WE ARE	STATUS
Labour reallocation & payback	1.5	H_reclaimed and dimensionless P_payback computed from sourced inputs	Scaffold prints them from typed constants.	NOT STARTED
Cost reduction	1	NCRI = 1 − κ, dimensionless	Not in either economics file.	NOT STARTED
Queuing model	1	M/M/1 or M/M/c with λ, μ, c, ρ stated	Not implemented.	NOT STARTED
Risk	1	FMEA ≥ 4 modes, RPN pre/post, mitigation shown in sim or code	Strong evidence available for “UAV–UAV conflict” (separation test 0.40 m → 3.29 m). Table not written.	NOT STARTED
Sensitivity	0.5	Graph on most uncertain input with break-even value	E077 branch has a sensitivity plot, but in currency and outside the graded script.	NOT STARTED

Component 3 — Studio Showcase & Viva Voce (5)

CRITERION	MK	REQUIRED EVIDENCE	WHERE WE ARE	STATUS
LinkedIn video (3–4 min)	1	Live sim, business graph, every member on webcam PiP	Simulation footage tooling ready (`render_scene_media.py`, live viewer). No recording.	NOT STARTED
AI audit & peer review	1	Three verified AI-error cases; review of Groups 06 and 11	None recorded.	NOT STARTED

CRITERION	MK	REQUIRED EVIDENCE	WHERE WE ARE	STATUS
Live Expo demo	1	Runs from fresh clone; FSM states shown; one handled failure	Live viewer works locally on macOS. Fresh-clone test not done; FSM overlay missing. Head-on conflict is a ready “handled failure”.	PARTIAL
Individual viva	2	Each member defends own role, code, and the whole project	E043 ready on flight/control. E051/E075 code regions empty. E077’s committed work is outside her role.	PARTIAL

4 Product Requirements & Status

Functional (FR) and non-functional (NFR) requirements of the product as it must exist on Saturday. Each row names the acceptance test that closes it.

ID	REQUIREMENT	ACCEPTANCE TEST	OWNER	STATUS
FR-01	MJCF model of the NMIMS Main Building (interior default, exterior variant) with two quadrotors	`mujoco.MjModel.from_xml_path(...)` loads; `src/test_env.py` passes	**E043**	DONE
FR-02	MJCF timestep 0.002 s (500 Hz) with friction and solref justified	Option reads 0.002; hover, patrol and separation test still pass; justification text in dossier §2.1	**E043**	NOT STARTED
FR-03	Cascaded closed-loop flight control with stated gains (units)	Gains table in dossier §2.1; altitude hold within ±0.01 m on the ground loop	**E043**	DONE (gains table pending)
FR-04	Coordinated dual-UAV patrol (sectors / offset ring) covering the corridor loop and atrium galleries	Both policies run; revisit and coverage metrics reported	**E043**	DONE
FR-05	Inter-UAV collision avoidance, $d_{min} \geq 2.5$ m	`tests/test_e043_separation.py` : layered method passes head-on	**E043**	DONE
FR-06	OpenCV HSV intruder detection on the onboard camera, bounding box + latency	`detect_intruder_hsv` returns bbox; IoU ≥ 0.70 vs segmentation ground truth at noon; latency logged	**E051**	NOT STARTED
FR-07	Inverse pinhole projection pixel → world (f = 450 px, 640 × 480)	Projected intruder position within stated error of true position	**E051**	NOT STARTED
FR-08	Restricted-zone geofence (ray-casting point-in-polygon), 4-tier alert with hysteresis, GPS/IMU noise injection, breach log	`evaluate_geofence_status` returns tiers 0–3 for test points; breach logged with timestamp	**E075**	NOT STARTED
FR-09	Surveillance FSM: PATROL → DETECT → VERIFY → ALERT(0–3) → TRACK → RETURN_TO_PATROL	State logged every row; state shown as overlay in the live viewer; one transition demonstrable live	**E043 integrates**	PARTIAL
FR-10	500 Hz telemetry: t, run_id, seed, pose, velocity, reference, tracking error, actuator forces, FSM state, events	One CSV per run; row interval 0.002 s	**E043 / E075 QA**	PARTIAL
FR-11	Per-run tracking RMSE and control effort $\int \tau^2 \, dt$	Columns present in `campus_patrol_real_trials.csv`	**E043**	NOT STARTED
FR-12	Monte Carlo runner: N runs per condition, randomised wind / intruder timing / start, seeds recorded	`analytics/campus_patrol_real_trials.csv` + provenance JSON	**E043**	PARTIAL

ID	REQUIREMENT	ACCEPTANCE TEST	OWNER	STATUS
FR-13	Statistics: Welch t-test, Cohen's d, 95 % CI, baseline vs proposed	`analytics/statistical_validation.py` prints t, df, p, d, CI from the CSV	**E075 / E043**	NOT STARTED
FR-14	Publication figures regenerate from real-trials CSV only	`generate_paper_figures.py` produces 3 figures with no watermark	**E043**	PARTIAL
FR-15	CSBS model: H_reclaimed, P_payback, T_payback, NCRI/κ, M/M/c, sensitivity ±30 %, break-even	Script re-run reproduces every number in dossier §3.1 from sourced inputs	**E077**	NOT STARTED
FR-16	FMEA ≥ 4 failure modes with RPN before/after mitigation	Each mitigation points to code or a simulation run	**E077 + all**	NOT STARTED
FR-17	Live viewer with FSM state and one handled failure	`mjpython src/view_live.py` shows states; head-on conflict resolved on screen	**E043**	PARTIAL
NFR-01	Runs from a fresh clone with `pip install -r requirements.txt`	Clean-directory clone on a second laptop (Windows or macOS)	**E043**	PARTIAL
NFR-02	Data provenance: no `SIMULATED_*` or `np.random` number reported as a result	Figure gate refuses unprovenanced CSV; dossier provenance boxes ticked honestly	**All**	DONE (gate)
NFR-03	Zero currency anywhere (code, dossier, video)	a case-insensitive grep of the deliverables for currency words, codes and symbols returns nothing	**E077**	VIOLATING
NFR-04	REP-103 frames and SI units (m, rad, m/s, N·m)	Stated in dossier §2	**E043**	DONE (MuJoCo ENU, SI)
NFR-05	Each member ≥ 4 conventional feature commits from own account on own branch	`git shortlog -sn -- Group_05_*/`	**All**	AT RISK
NFR-06	No videos, venvs or large binaries in the monorepo	MP4s stay out of the PR; video lives on LinkedIn	**E043**	PARTIAL (two MP4s in our repo)

5 What Is Done (Evidence)

AREA	DELIVERED	EVIDENCE IN THE REPO
Repository	Course scaffold layout preserved; pinned `requirements.txt`; README onboarding; private GitHub repo with `main` owned by E043	`README.md`, `requirements.txt`, commit `c3fdea7`
Scene	NMIMS Main Building interior: 10 floors × 3.6 m, central atrium 21.6 × 38.5 m (satellite-measured), 132 m ground-floor corridor loop, galleries, VIP Lounge (restricted zone), cafeteria, lift lobby, Wall of Fame; exterior campus variant from OpenStreetMap	`models/nmims_main_interior.xml`, `models/scene_source/*`, `interior_manifest.csv` (MEASURED vs ASSUMED)
Vehicles	Two 1.35 kg "+" quadrotors, arm 0.26 m, four rotors 0–7.5 N each with yaw drag torque, nadir and forward cameras (640 × 480, f = 450 px), IMU sensors	`models/uav_fleet.xml`
Control law	Position P → velocity PI → flat-output inversion → attitude PD with gyroscopic feed-forward; exact inverse mixer; tilt slew limiter; principal-frame inertia	`src/aerial_patrol_swarm.py` (E043 region)
Mission	Phases takeoff → transit → descend → patrol; policies `sectors` and `offset_ring`; per-UAV yaw modes	same
Safety	Altitude layers + closing-speed governor + right-hand sidestep on top of the guide's APF	`tests/test_e043_separation.py`: APF alone 0.40 m (contact) → layered 3.29 m (pass)
Disturbance	Ornstein–Uhlenbeck gust field written to MuJoCo wind	`_gust_step`
Metrics	Revisit intervals, gallery coverage, energy (Wh), collisions, altitude error	console summary + telemetry CSV
Analytics	Seeded batch runner with provenance (seeds, code/model hashes, versions); figure generator that refuses unprovenanced data	`analytics/run_benchmark.py`, `generate_paper_figures.py`
Team tooling	Region-ownership PR gate; live viewer; stills + 2 × 2 patrol video renderer	`src/check_region_ownership.py`, `view_live.py`, `render_scene_media.py`
Planning	PRD v2.0 and Implementation Plan v2.0 (exterior scene) with Mermaid sources	`docs/planning/`

DEVELOPMENT CHECKS, NOT RESULTS. Single 240 s interior runs (one seed) gave: `sectors` worst ground-loop revisit 53.4 s, 66.7 % gallery coverage, 0 collisions; `offset_ring` 36.7 s and 0 % gallery coverage. These show the pipeline works. They are **NOT** Monte Carlo evidence and must not appear in the dossier as results.

Work on other branches (reviewed, not merged)

E077's `feat/e077-guardian-live-surveillance-upgrade` (10 commits, 381 files, ≈ 86 k lines) adds a YOLO/ByteTrack runtime, A* navigation, a control-room page and a currency-denominated economics module under `integrations/`. It is carefully caveated but: fails the ownership gate on all files, leaves `campus_security_economics.py` untouched, uses currency, is Windows/CUDA-tested only, and 8 of 10 commits carry a mistyped e-mail (`@mgail.com`) so GitHub does not attribute them to her account. Recommendation: park it; salvage only the "every assumption needs a provenance string" idea.

6 What Is Left — Master Checklist (Prioritised)

P0 — submission is invalid or a component scores zero without it

ID	TASK	OWNER	DONE WHEN
L-01	Tell the course faculty the dossier is late; agree a turn-in time; raise §15 questions	**E043**	Written reply received
L-02	Copy the official `MDRIIA_Group_05_Technical_Dossier.docx`, `docs/README.md`, `docs/figures/README.md` from the monorepo into our `docs/`	**E043**	Files present, unmodified
L-03	Rename scaffold data to `analytics/SIMULATED_campus_patrol_benchmark.csv`; point runner output to `analytics/campus_patrol_real_trials.csv`	**E043**	Figure script reads only the real-trials file
L-04	Rewrite `campus_security_economics.py`: delete every hard-coded result; compute from sourced inputs and the real-trials CSV; ratios only	**E077**	No literal outputs; `grep` finds no currency
L-05	Run the Monte Carlo batch (baseline + proposed, N ≥ 30 each, seeds recorded)	**E043**	CSV + provenance JSON committed
L-06	Fill all six dossier sections and sign-off in Word desktop; export PDF; commit both	**All · E043 compiles**	.docx + .pdf in `docs/`
L-07	Each of the four members turns in the PDF on MS Teams	**Each**	Four “Turned in” confirmations
L-08	Open the PR of our group folder into the course monorepo (`github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026`), touching only `Group_05_…/`	**E043**	PR open, checks green, no MP4/venv
L-09	Fill §1.1 claim table: verify or Correct/Delete “72 % latency”, “11-month CapEx recovery”, “90-s verification”; also drop the README/roster “45 → 8.5 min” unless measured	**E077 + E043**	Each row has source type + verdict
L-10	Declare all AI use (§5.1 three cases, §6 AI-drafted share per member)	**All**	Honest entries signed in sign-off

P1 — needed for full marks in a criterion

ID	TASK	OWNER	DONE WHEN
L-11	Set timestep 0.002 s; re-check hover, patrol, separation test; write friction/solref justification	**E043**	Tests pass at 0.002 s
L-12	500 Hz per-run telemetry with reference, tracking error, actuator forces, FSM state	**E043**	Row interval = 0.002 s
L-13	Per-run tracking RMSE and $\int \tau^2\,dt$ in the trials CSV	**E043**	Two new columns
L-14	Implement `detect_intruder_hsv` + `inverse_pinhole_projection`; measure IoU and latency, noon vs twilight	**E051**	`vision_illumination_benchmark.csv` written
L-15	Implement `evaluate_geofence_status` (ray casting, 4 tiers, hysteresis, noise) + breach log	**E075**	Unit cases for tiers 0–3 pass
L-16	Wire the surveillance FSM; log state per row; overlay in viewer	**E043**	One live DETECT→ALERT transition
L-17	`analytics/statistical_validation.py`: Welch t, df, p, Cohen’s d, 95 % CI	**E075**	Numbers match dossier §4 table

ID	TASK	OWNER	DONE WHEN
L–18	Regenerate the three figures from real data; replace starter PNGs	**E043**	No watermark
L–19	Business: H_reclaimed, P_payback/T_payback, NCRI/κ, M/M/c (λ, μ, c, ρ), sensitivity ±30 % + break-even, queue graph	**E077**	Two graphs + §3.1 table reproducible
L–20	FMEA ≥ 4 modes, RPN pre/post, each mitigation shown in sim/code	**E077 + all**	Table in §3.2
L–21	Record and post the 3–4 min LinkedIn video per Showcase Guide §7; add URL to dossier §5 and `TEAM_ROSTER.json`	**All · E043 edits**	Live URL
L–22	Peer-review Group 06 and Group 11 videos	**All**	§5.2 filled
L–23	Standards decisions §1.2 (ISO 10218, ISO 3691-4, ANSI/RIA R15.06, IEC 60601-1, India Drone Rules 2021) + ISO 21384 / ASTM F3322 from the Standards Manual	**E075 + E043**	Yes/No + clause or reason for each
L–24	Baseline friction §1.3: today's guard patrol metric with a primary source	**E077**	Value + unit + citation

P2 — quality and viva safety

ID	TASK	OWNER	DONE WHEN
L–25	Fresh-clone rehearsal on a second laptop; fix anything missing	**E043**	Viewer runs from scratch
L–26	Reconcile SAP IDs: dossier template lists 70362400058 / …040 / 70362300055 / 70362300043; `TEAM_ROSTER.json` and README list …030 / …067 / …003 / …080	**All**	One set, confirmed with the official list
L–27	E077: fix git e-mail, re-author commits, move economics to `feat/e077-csbs-campus-security`	**E077**	Commits show as Subhankar-Uro
L–28	Assigned branch names for all four members	**All**	Branches exist with ≥ 4 commits each
L–29	Update PRD/Implementation Plan to the interior scene (v2.1) — optional	**E043**	—
L–30	Each member rehearses the five standard viva questions (§14)	**All**	30-second whiteboard rule met

Dependency chain

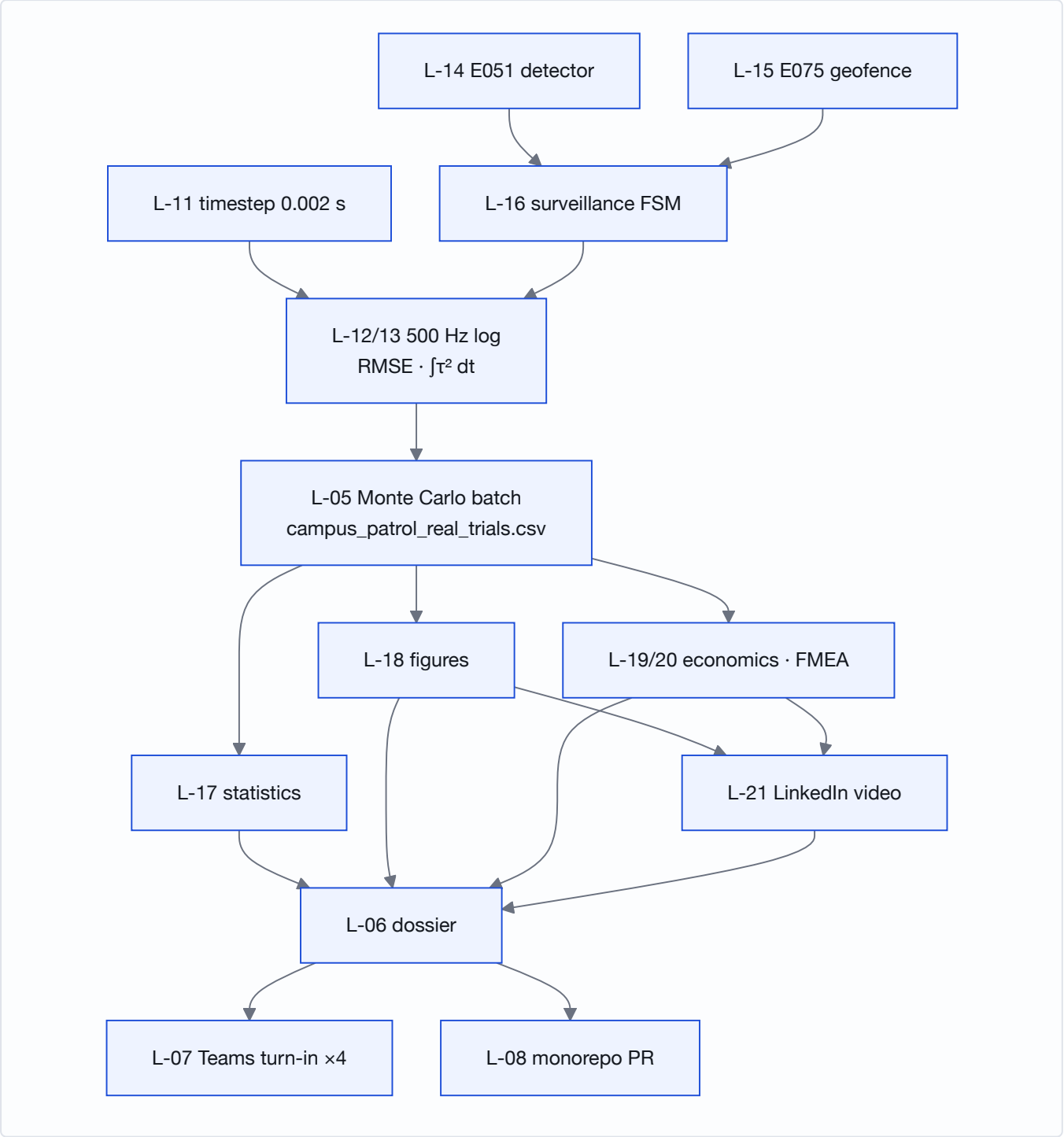


Figure 1 — Critical path. The Monte Carlo batch is the bottleneck: statistics, figures, economics and the video all consume it. If E051/E075 slip, run the batch with detection disabled and report the cycle-time comparison first (see §9 fallback).

7 Work Packages per Member

E043 Muaaz — Lead, flight dynamics, integration, owner of `main`

M–1	Timestep 0.002 s; re-tune if needed; justify friction 1.0/0.01/0.005 and solref 0.004 1 (§2.1)
M–2	500 Hz telemetry columns; per-run RMSE vs waypoint path; $\int \tau^2 \, dt$ over the eight rotor actuators
M–3	Surveillance FSM integration + viewer overlay; merge E051/E075 regions after the ownership gate
M–4	Runner → `campus_patrol_real_trials.csv`; baseline and proposed conditions; N per §15 ruling
M–5	Repo format (L-02, L-03), monorepo PR (L-08), fresh-clone test (L-25)
M–6	Dossier §2: settings table, gains with units, FSM list, authorship table, ≤ 25-line snippet, FBD worked example (hover: m·g = 1.35 × 9.81 = 13.24 N ⇒ 3.31 N per rotor; max 4 × 7.5 = 30 N ⇒ thrust-to-weight 2.27)
M–7	Video edit and post; compile and export the dossier

E051 Aditya — Computer vision (OpenCV), detection and tracking

A–1	`detect_intruder_hsv`: HSV threshold → morphology → contours → largest bbox; return detected, bbox, latency (ms)
A–2	`inverse_pinhole_projection`: (u, v) + camera pose + depth → world point; report error vs true intruder position
A–3	IoU vs MuJoCo segmentation mask (target ≥ 0.70) and latency at noon vs twilight lighting → `analytics/vision_illumination_benchmark.csv`
A–4	FMEA row “missed intruder at twilight” with measured miss rate; AI-audit case from own work
A–5	≥ 4 commits on `feat/e051-computer-vision-open`, own account

E075 Zaid — Geofencing, telemetry QA (owner of dossier §4)

Z–1	`evaluate_geofence_status`: ray-casting point-in-polygon on the VIP Lounge zone; tiers 0–3 with hysteresis
Z–2	GPS/IMU noise injection (stated σ); JSON breach log with timestamps
Z–3	Optional handled failure for the demo: return-to-home after ≥ 2.5 s telemetry loss (ISO 21384 / ASTM F3322 per Standards Manual §2.4)
Z–4	QA of the batch: logging rate, seeds, N; `statistical_validation.py`; fill §4 table
Z–5	≥ 4 commits on `feat/e075-restricted-zone-geo-`, own account

E077 Soumya — CSBS business and techno-economics (owner of dossier §3)

S–1	Rewrite `campus_security_economics.py` (see §10); delete constants 48.72, 7.84, 72.0, 50.4, 0.66, 0.31, 35.8, 1.2
S–2	Sourced guard-patrol inputs (walking speed, shift pattern, rounds per shift) or labelled assumptions
S–3	Sensitivity and queue graphs (PNG, 300 DPI) for the dossier and the video
S–4	FMEA table; §1.1 claim verification; §1.3 baseline friction
S–5	Fix commit e-mail; work on `feat/e077-csbs-campus-security`; park the runtime branch

8 Technical Dossier — Field-by-Field Fill Plan

Every field in the official template, who fills it, and where the value comes from. Fill in Word desktop; never delete a "Select ▾" or "type here" control.

SECTION · FIELD	OWNER	SOURCE OF THE VALUE	STATUS
Cover · roster check	All	Resolve SAP ID mismatch (L-26)	DECISION
§1.1 Claim verification (3 rows)	E077, E043	Primary sources or Correct/Delete; simulation measurement	NOT STARTED
§1.2 Standards (5 rows)	E075, E043	Faculty hints: four "No", India Drone Rules "check"; add ISO 21384 / ASTM F3322	NOT STARTED
§1.3 Baseline friction	E077	Published guard-patrol metric with citation	NOT STARTED
§2.1 Timestep, friction, solref, law, gains, FSM, loads?	E043	Final MJCF + controller constants	PARTIAL
§2.2 Authorship table	All	Git history; scaffold files = Modified, generators = Written	NOT STARTED
§2.3 Code snippet ≤ 25 lines (text, Consolas 8 pt)	E043	FSM transition or attitude law with line numbers	NOT STARTED
§2.4 FBD + worked equation	E043	Hover-thrust example (§7 M-6), telemetry column reference	NOT STARTED
§3.1 Business table, queue model, λ μ c, T_payback, reproducible?	E077	Rewritten script	NOT STARTED
§3.2 FMEA (2 seeded + ≥ 2 more) + two graphs	E077 + all	Separation test, vision benchmark, geofence log	NOT STARTED
§4 Protocol: baseline, N, distributions, Hz, seeds, CSV path	E075	Runner provenance JSON	NOT STARTED
§4 Results table: RMSE, $\int \tau^2\,dt$, primary metric, test, t/df/p, d, CI	E075	`statistical_validation.py`	NOT STARTED
§4 Trajectory plot + box-plot	E043	Regenerated figures	NOT STARTED
§4 Data Provenance Declaration (4 boxes)	All	Tick only what is true	NOT STARTED
§5 Video URL, length, three yes/no checks	E043	LinkedIn post	NOT STARTED
§5.1 AI audit AI-01…AI-03	All	Real cases caught by docs/paper/running code	NOT STARTED
§5.2 Peer review: Group 06 (Crawler Solar Cleaner), Group 11 (Solar PV Inspection AMR)	All	Their LinkedIn videos	NOT STARTED
§6 Learning record (hours, role match, commit share, AI share, …)	Each member own row	`git shortlog -sn -- Group_05_*/`	NOT STARTED
Sign-off (original, code runs, viva-ready, signature, consent)	Each member	Personal	NOT STARTED

TEMPLATE FACTS TO CORRECT IN §2. The pre-filled text says "9 actuators (2 × 4 rotors + 1)" and "PATROL (Alpha NE / Bravo SW sectors)". Our model has **8** rotor actuators and the interior patrol is Alpha on the ground-floor loop and Bravo on the atrium galleries. State the real values in the "type here" fields.

9 Experimental Design Still to Run

ITEM	SPECIFICATION
Proposed condition	Dual-UAV, `sectors` policy, interior scene, surveillance FSM active
Baseline condition (primary)	Modelled static-guard foot patrol of the same 132 m loop and galleries, walking speed and stops from a cited source — matches the title (“compared to static security guard patrols”)
Secondary comparison	`offset_ring` vs `sectors` (both already implemented)
N	≥ 30 per condition (rubric); ≥ 50 recommended to also satisfy the Standards Manual §4 — ask sir (§15)
Randomised inputs	Mean wind and gust σ (uniform range, stated), intruder appearance time, initial dock jitter; one seed per run, recorded
Logging	500 Hz, one CSV per run (gitignored) + one summary row per run in `campus_patrol_real_trials.csv`
Measures	Patrol cycle time / worst revisit (primary), detection latency (s), IoU, inter-UAV d_{min}, tracking RMSE, $\int \tau^2 dt$, energy Wh/km
Statistics	Welch two-sided t-test $\alpha = 0.05$, df, exact p, Cohen’s d, 95 % CI of the difference
Run time budget	Batch length should be measured on the actual laptop before committing to 900 s runs; shorten runs (e.g. ≥ 2 full ground laps) if N × duration does not fit before Saturday — and say so in §4
Fallback if E051/E075 slip	Run cycle-time and separation trials with detection disabled; report detection latency as “not measured” rather than estimating it

10 CSBS Business Model Still to Build

All quantities dimensionless or in hours; no currency anywhere. Every input is either sourced (author, year, DOI/URL) or labelled *assumption* and swept in the sensitivity graph.

QUANTITY	DEFINITION (AS IN THE DOSSIER)	INPUTS FROM
H_reclaimed	Guard-hours per day moved from walking rounds to incident response	Shift pattern (sourced/assumed) × fraction of rounds replaced (from measured cycle times)
P_payback	CapEx / (T_life · ΔOpEx_annual); P < 1 ⇒ pays back within life	CapEx parity factor, T_life, ΔOpEx — all as ratios to the guard baseline
T_payback	CapEx / ΔOpEx, in months	Same
NCRI, κ	NCRI = (C_baseline − C_robot) / C_baseline; κ = 1 − NCRI (manual target κ ≤ 0.25)	Normalised cost components
Queue	M/M/c: λ incidents/h, μ verifications/h per responder, c responders; ρ = λ/(cμ) < 1; W_q via Erlang C	λ assumption; μ from simulated verification time
ΔFTE (manual)	ΔFTE = annual reclaimed hours / 2080	H_reclaimed × 365
Sensitivity	Most uncertain input ±30 %; plot P_payback and NCRI; mark break-even	Script sweep

FMEA seed list (≥ 4 required)

FAILURE MODE	MITIGATION ALREADY IN, OR TO ADD	EVIDENCE
UAV–UAV conflict at crossing (d < 2.5 m) — seeded	Altitude layers + closing-speed governor + sidestep	Separation test 0.40 m → 3.29 m
Missed intruder at twilight (low IoU) — seeded	HSV range / exposure tuning (E051)	Vision illumination benchmark
Wind gust drives UAV into wall	Tilt limiter, velocity PI, corridor routing	Collision counter = 0 across trials
Telemetry / link loss	Return-to-home after ≥ 2.5 s (E075, optional)	Demo run
False geofence alarm from GPS noise	Hysteresis on tier changes (E075)	Breach log with noise on/off

11 Repository & Git Compliance

REQUIRED BY SIR	OUR REPO NOW	ACTION	STATUS
`analytics/campus_patrol_real_trials.csv`	Absent	L-03, L-05	MISSING
`analytics/SIMULATED_campus_patrol_benchmark.csv`	Named `campus_patrol_benchmark.csv`	Rename (L-03)	MISSING
`docs/MDRIIA_Group_05_Technical_Dossier.docx` + `.pdf`	Absent	L-02, L-06	MISSING
`docs/README.md` , `docs/figures/README.md`	Absent	Copy from monorepo	MISSING
`models/campus_perimeter_patrol.xml` , `src/aerial_patrol_swarm.py` , `src/test_env.py` , `analytics/campus_security_economics.py` , `generate_paper_figures.py`	Present (names preserved)	—	DONE
Extra files (scene generators, viewer, tests, planning)	Present	Allowed inside the group folder; keep MP4s out of the PR	CHECK
Branches `feat/e043-lead-uav-flight-dyna` , `feat/e051-computer-vision-open` , `feat/e075-restricted-zone-geo-` , `feat/e077-csbs-campus-security`	Only `main` + five non-standard e077 branches	L-28	MISSING
Conventional commits, ≥ 4 per student, own account	1 commit (E043); E077 commits mis-attributed	L-27, L-28	AT RISK
Work submitted to the course monorepo via PR into `main`	Only in private `MuaazSM/mdriia-project`	L-08	MISSING
No direct commits to `main` (Standards Manual §5)	Initial commit made directly to our private `main`	All further work via PRs	NOTE

12 Integrity Risks & Blockers

#	RISK	IMPACT	RESPONSE
R1	Dossier and video deadlines passed	Possible late penalty or no marks	Contact sir tonight (L-01); submit as early as possible
R2	Scaffold `np.random` numbers or the README's "45 → 8.5 min" reused as results	Zero for the component (misconduct)	Figure gate; rename to `SIMULATED_`; quote only real-trials numbers
R3	Hard-coded economics constants	C2 zero	L-04
R4	Currency in E077's work	Zero Currency Rule breach	Ratios only
R5	Undeclared AI use	Integrity breach	Full §5.1 / §6 disclosure
R6	Attribution mismatch: work in git ≠ declared role; commits from wrong e-mail	Viva marks; integrity question	Each member commits own region from own account; honest §2.2 and §6
R7	E051/E075 regions still empty two days before viva	C1 control + two members' viva	Start tonight; E043 reviews same day
R8	Batch runtime too long for N ≥ 30–50 at 900 s each	No statistics	Measure one run; shorten duration with justification
R9	Indoor scene vs "perimeter patrol" title	Problem-framing / mechanism-match marks	Ask sir (§15); exterior variant ready as fallback
R10	YOLO (AGPL-3.0, PyTorch) instead of OpenCV	Changes the research question	Keep OpenCV HSV for the graded work
R11	Disk space on the lead laptop is nearly full (≈ 0.3–0.5 GB free)	Batch runs / video export may fail	Free space before running the batch and recording

13 Plan to Saturday 31 October

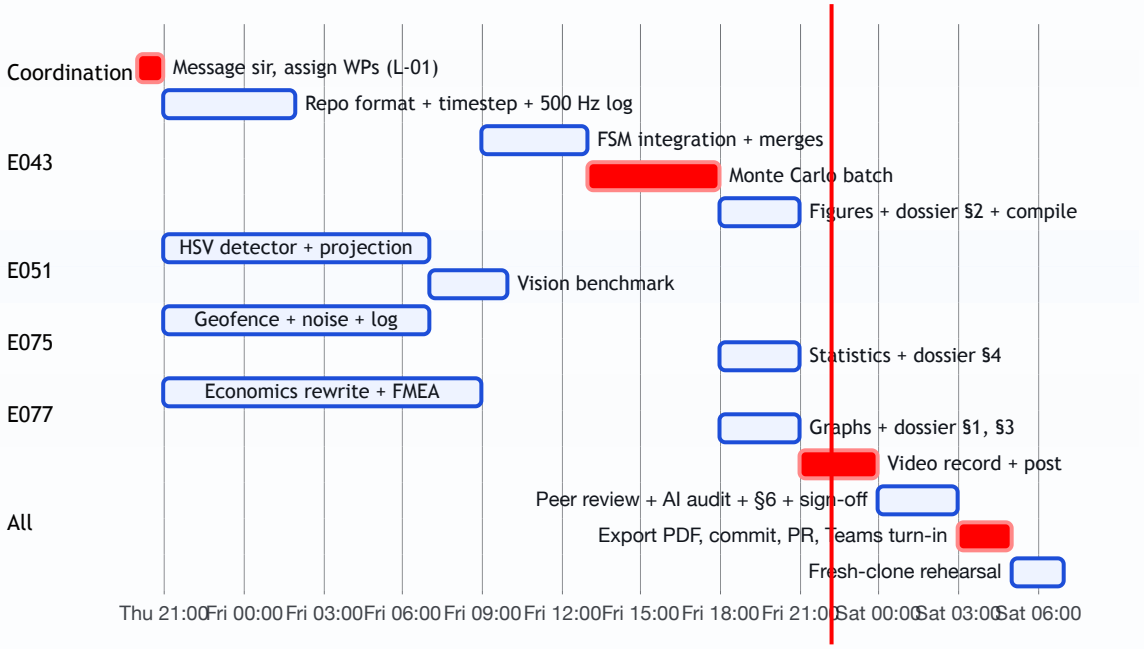


Figure 2 — Compressed schedule. Start times are targets; the batch (M-4) gates the statistics, figures, economics graphs and video. If it slips, record the video with the cycle-time results only.

14 Viva & Expo Readiness

QUESTION SIR WILL ASK (SHOWCASE GUIDE §11)	WHO ANSWERS	PREPARED ANSWER SOURCE
Why that solref time constant and friction? What happens if you halve them?	E043	§2.1 justification; try it live in the viewer
Show one FSM transition live and the line of code that triggers it	E043 (+ E051 DETECT, E075 ALERT)	Viewer overlay + §2.3 snippet
Which input moves payback most, and where does its value come from?	E077	Sensitivity graph + source
Why Welch's t-test? What does Cohen's d mean in plain words?	E075	Unequal variances; d = gap in pooled-SD units
What did AI get wrong, and how did you prove it?	Each member	§5.1 cases

Expo checklist

- [ ] Laptop with simulation running from a fresh `git clone` + `pip install -r requirements.txt`
- [ ] FSM states visible in the viewer; one handled failure (head-on UAV conflict resolved by the governor and sidestep)
- [ ] Business sensitivity and queue graphs open
- [ ] Webcam picture-in-picture of every member; no PPT slides
- [ ] Each member can explain their code in 30 seconds on a whiteboard

15 Decisions Needed from the Course Faculty

#	QUESTION	OUR PROPOSAL
Q1	Will a late dossier (after 26 Oct) be accepted, and by when?	Turn in by Saturday morning before the Expo
Q2	Is an **indoor** patrol of the NMIMS Main Building acceptable under the "campus perimeter patrol" title?	Keep indoor as primary (corridor loop = guard round); exterior perimeter variant available
Q3	N ≥ 30 (rubric) or N ≥ 50 (Standards Manual)?	Target 50 if run time allows, minimum 30
Q4	Timestep 0.002 s mandatory, or 0.004 s with honest 250 Hz reporting?	Move to 0.002 s
Q5	May detection use YOLO, or must it stay OpenCV as in the title?	OpenCV HSV for graded work
Q6	Correct SAP IDs: dossier template vs `TEAM_ROSTER.json` disagree for all four members	Use the official list; fix the other file

Prepared by E043 for Group 05. Status reflects the repository at commit `c3fdea7` (main) and branch `feat/e077-guardian-live-surveillance-upgrade` at `be80886`, and the course monorepo as read on 29 Oct 2026.